How-To Library

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Enhancing Reference 360 UI Performance for Large Data Sets

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Abstract

This article explains how to improve loading time of Reference 360 assets, such as code lists and hierarchies, in the user interface when you use a very large volume of reference data.

Supported Versions

- MDM - Reference 360 July 2024 or later

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Overview

When you use a very large volume of reference data, the loading time of the Reference 360 assets, such as code lists and hierarchies, in the user interface might take longer than expected.

To reduce the loading time of these assets, optimize them at least once for your organization. If you add or update assets, you can optimize the incremental assets to improve their loading time.

Note: The loading time and performance improvement might depend on your environment and network latency.

When you optimize assets, an asset optimization job runs. You can monitor the job on the **My Jobs** page.

To troubleshoot any issues with asset optimization, contact Informatica Global Customer Support.

You can optimize the following types of assets:

- Code lists
- Hierarchies

Note: You can't optimize crosswalks.

Prerequisites to optimize assets

Before you optimize assets in a production environment, ensure that you complete the following prerequisites:

- Enable the asset optimization option in the user interface by running the asset optimization modeling REST API.
  For more information about the asset optimization modeling REST API, see "Enable asset optimization by using REST API" on page 3.

- Optimize assets in your development and QA environments to assess the performance impact of asset optimization.

Enable asset optimization by using REST API

To enable the asset optimization option in the user interface, use the asset optimization modeling REST API . You can also retrieve the status of an asset optimization modeling job.

Enable the asset optimization option

By default, the asset optimization option isn't available in the user interface. You must enable it by running the asset optimization modeling REST API. The REST API also analyzes the number of assets to be optimized and sets them up for optimization. When you run the asset optimization modeling REST API, it triggers an asset optimization modeling job.

Enabling the asset optimization option is a one-time activity for an organization.

POST request

To enable the asset optimization option and to run an asset optimization modeling job, submit a POST request with the following URI:

```
/rdm-service/external/v1/asset/optimization/modelling
```

POST response

The response is in the JSON format.

The following table describes the attributes in the response:

Field	Type	Description
jobId	String	ID of the asset optimization modeling job.
type	String	Type of the job, such as asset optimization modeling.
status	String	Status of the asset optimization modeling job. Value can be `RUNNING`, `SUCCESS`, `FAILED`, or `CANCELLED`.
createdBy	String	User name of the user who triggered the asset optimization modeling job.
createdDate	String	Date when the user ran the asset optimization modeling job.

POST example

To enable the asset optimization option, you might use the following request:

```
{
   "method":"POST",
   "url":"https://qa-pod1-mdm.mrel.infaqa.com/rdm-service/external/v1/asset/optimization/
modelling",
   "headers":{
      "accept":"application/json",
      "Session ID":"XXXXXXXXXXXXXXXXXXXXXX"
   },
   "data":""
}
```

The following sample response shows the details of an asset optimization modeling report:

```
{
  "jobId": "663a68d9ec6cff7ea71b48ed",
  "type": "ASSET_OPTIMIZATION_MODELLING",
  "status": "RUNNING",
  "createdBy": "cLuE2VONraVl6EjvWO9Hw5",
  "createdDate": "2024-05-07T17:46:01.768+00:00"
}
```

Get asset modeling job status

Retrieves the status of an asset optimization modeling job and the number of assets that the job processed..

GET request

To get the status of an asset optimization modeling job, submit a GET request with the following URI:

```
/rdm-service/external/v1/asset/optimization/modelling/job/{jobId}
```

GET response

The response contains the details of an asset optimization modeling job, such as status of the job, created date, and number of code lists and hierarchies that can be optimized.

The following table describes the attributes in the response:

Field	Type	Description
jobId	String	ID of the asset optimization modeling job.
type	String	Type of the job, such as asset optimization modeling.
status	String	Status of the asset optimization modeling job. Values can be `RUNNING`, `SUCCESS`, `FAILED`, or `CANCELLED`.
createdBy	String	User name of the user who triggered the asset optimization modeling job.
createdDate	String	Date when the user ran the asset optimization modeling job.
detail	-	Details of a job at a specific time.
numberOfCodelistModelled	Integer	Number of code lists that are ready to be optimized.
numberOfHierarchyModelled	Integer	Number of hierarchies that are ready to be optimized.
errors	-	Details of the error that the REST API request returns.
errorCode	String	Error code for the error type.
errorSummary	String	Message that explains why the invalid records couldn't be set up for optimization.
errorParameter	String	Parameter that provides details of an error.
errorCauses	String	Causes of the error to explain why the assets couldn't be set up for optimization.

GET example

To get the status of an asset optimization modeling job, you might use the following request:

```
{
    "method": "GET",
    "url": "https://use4-mdm.dm-us.informaticacloud.com/rdm-service/external/v1/asset/
optimization/modelling/job/97d517fc22a78aea89646b1na",
    "protocol": "HTTP/1.1",
    "headers": {
        "IDS-SESSION-ID": "XXXXXXXXXXXXXXXXXXXXXX"
    }
}
```

The following sample response shows the status of an asset optimization modeling job:

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Content-Length: 193

{
   "jobId":"663a68d9ec6cff7ea71b48ed",
   "type":"ASSET_OPTIMIZATION_MODELLING",
   "status":"SUCCESS",
   "createdBy":"cLuE2VONraVl6EjvWO9Hw5",
   "createdDate":"2024-05-07T17:46:01.768+00:00",
   "detail":{
      "numberOfCodelistModelled":0,
      "numberOfHierarchyModelled":0
   }
}
```

Optimize assets

After you enable the asset optimization option, you can optimize assets to improve the loading time of code lists or hierarchies in the user interface.

1. In the **Explore** panel, click **Actions** > **Optimize asset performance**.

   Reference 360 displays the number of assets that require optimization.
2. In the **Confirm Asset Optimization** dialog box, click **Optimize Assets**.

   An asset optimization job starts. You can monitor the job on the **My Jobs** page. To view the job steps, see "Monitor asset optimization jobs" on page 5.

Monitor asset optimization jobs

An asset optimization job optimizes the code lists and hierarchies to improve their loading time in the user interface.

After you trigger an asset optimization job, you can monitor and view the job details on the **My Jobs** page.

An asset optimization job includes the following steps:

1. Lock the Job
2. Optimize Assets
3. Release the Job

Lock the Job

The lock the job step locks the job instance so that you can run only a single job instance at a time.

Optimize Assets

The optimize assets step processes the code lists and hierarchies in your organization for optimization.

After the system locks the job instance, the optimize assets step processes the assets to improve the loading time.

Note: If your organization contains hierarchies with more than 100 relationships, the optimize assets step processes the first 100 relationships and ignores the remaining relationships.

The optimize assets step updates the job metrics with the total number of records processed. You can also select an asset and view the step metrics for the selected asset.

The following table describes the metrics for the optimize assets step:

Metric	Description
Total Records	Number of records that the optimize assets step processed.
Error Records	Number of records that the step didn't process because of internal errors.
Success Records	Number of records that are optimized and are moved to the release the job step.

The following table describes the metrics that you can view for the optimize assets step for an asset you select:

Metric	Description
Starting Records for Asset	Number of records that the optimize assets step processed for the selected asset.
Error Records	Number of records that the step didn't process because of internal errors for the selected asset.
Success Records	Number of records that are optimized and are moved to the release the job step for the selected asset.

Release the Job

The release the job step releases the locked job instance.

Author

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